LECTURE NOTES IN NETWORKS AND SYSTEMS

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Hybrid Model for Detecting Alcoholism from EEG Signals using Deep Learning and Machine Learning

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I. INTRODUCTION

Problematic Reality

In the World

Alcoholism is a global health crisis responsible for 3 million annual deaths (5.3% globally) and is associated with over 200 diseases, notably cirrhosis and cardiovascular conditions (WHO, 2024).

In the Americas

In Latin America, alcoholism is a severe public health crisis responsible for nearly 85,000 annual deaths, with alcohol consumption rates exceeding the global average (PAHO, 2024).

In Peru

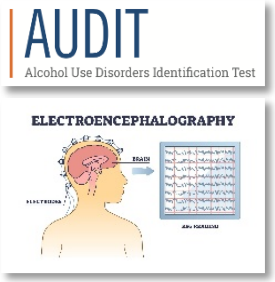
In Peru, alcoholism is a major public health crisis affecting 1.5 to 2 million adults and contributing to thousands of deaths each yea

I. INTRODUCTION

Current Situation

Medical consultation

Tests to diagnose alcoholism



Evaluation of results and diagnosis

Issues

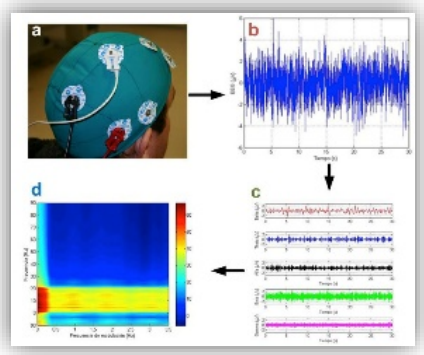
Interpreting the results is time-consuming

Limited availability of medical specialists

High diagnostic costs

I. INTRODUCTION

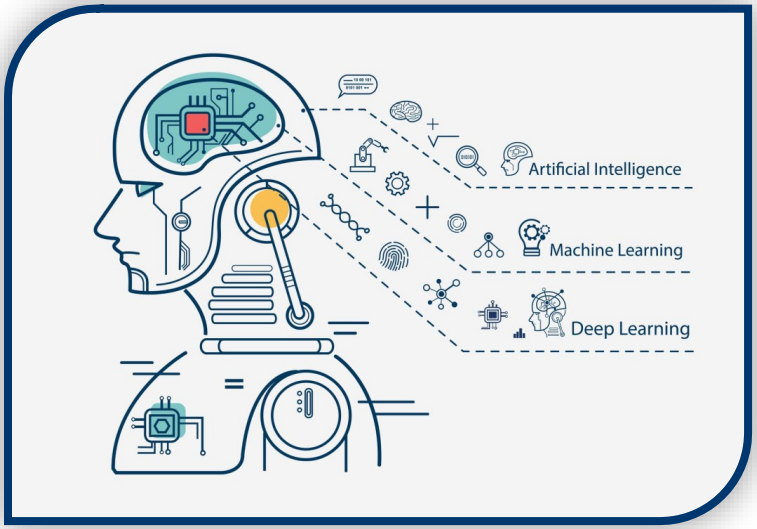
Benefits



Analysis of aggregated clinical data from patients using brain signals

Proposed Solution

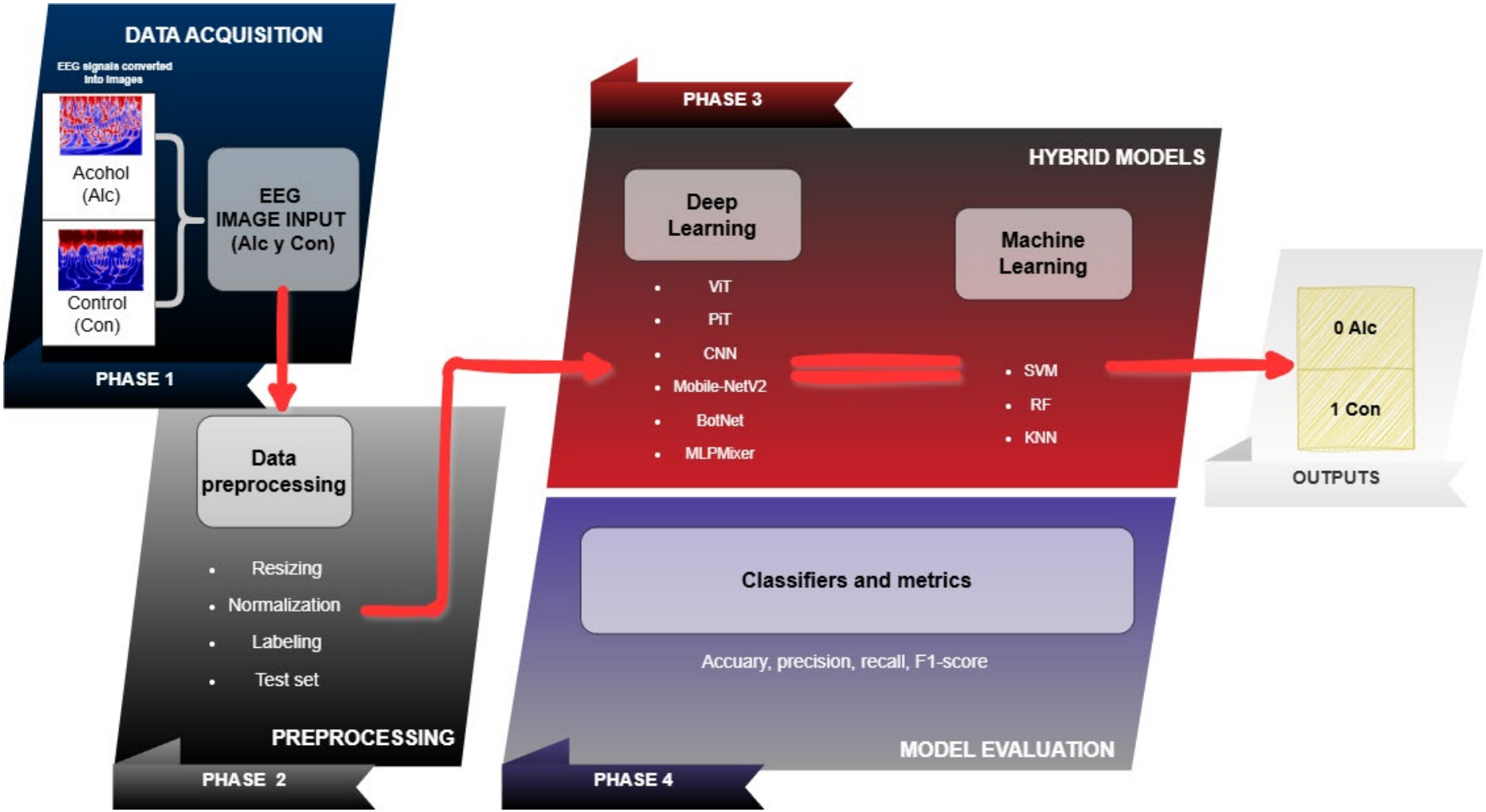
Evaluation for the detection of alcoholism using EEG signals



Hybrid Model for Detecting Alcoholism from EEG Signals using Deep Learning and Machine Learning

More accurate detection of alcoholism and early diagnosis

II. METHODOLOGY



PHASE 1: DATASET ACQUISITION

Source: Kaggle – Wavelet-transformed EEG data for alcohol detection.

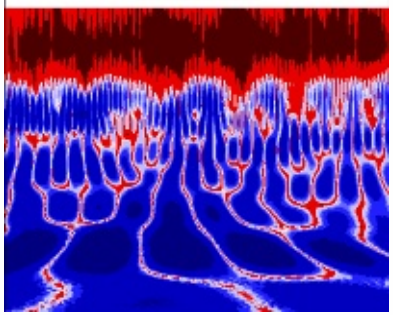
Signal Origin: 61 brain channels (EEG electrodes).

Transformation: Continuous Wavelet Transform (CWT) to create Scalograms.

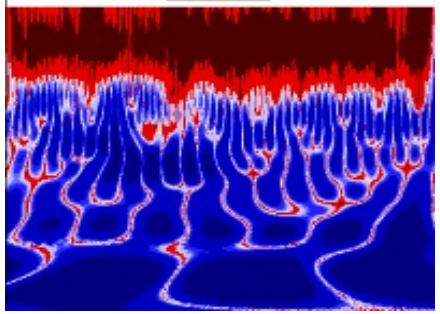
kaggle

Total Images: 924	**Alcohol Subjects (Alc):** 468
Control Subjects (Con): 456	**Original Resolution:** 200 x 172 pixels (Time x Frequency).

Alcoholic



Control



Fuente: **Kaggle (Wavelet-Transformed EEG Data for Alcohol Detection)**

PHASE 2: PREPROCESSING

Resizing

Original: 200x172 px

Target: **224x224px**

Standard input for Deep Lerning architectures

Normalization

Pixel Values: 0-255

Range: **[0, 1]**

Ensures numerical stability

Labeling

Control **(Con)**

Alcohol **(Alc)**

Test set

80% Training (739 images)

20% Testing (185 images)
Reserver for metrics

PHASE 3: HYBRIDS MODELS

In terms of methodology, six deep learning algorithms and three machine learning algorithms were used. Hyperparameter tuning was applied to all of them.

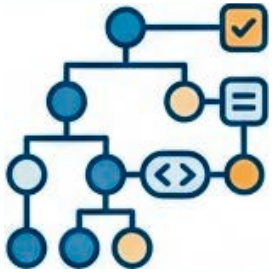
The Extractor (Deep Learning)

Responsible for “understanding” the image and extracting complex features.

- **ViT**
- **PiT**
- **CNN**
- **MobileNetV2**
- **BotNet**
- **MLPMixer**

Feature vectors

The Classifier (Machine Learning)



Responsible for making the final binary decision based on the extracted data.

- **SVM (Support Vector Machine)**
- **Random Forest (RF)**
- **K-NN**

PHASE 4: MODEL EVALUATION

- The data was divided into an 80% training set and a 20% test set.
- Electroencephalogram images were preprocessed and then evaluated using hybrid models.
- Six deep learning models and three machine learning models were used to perform the evaluation with hyperparameter tuning.
- The following metrics were used for this evaluation: accuracy, recall, precision, F1 score, confusion matrix, and ROC curve.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

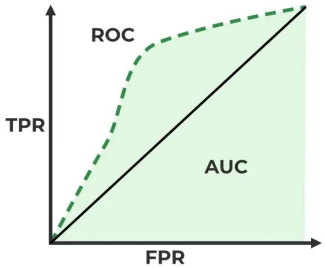
$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = \frac{2(\text{Recall} * \text{Precision})}{\text{Recall} + \text{Precision}}$$

Confusion Matrix

ROC Curve



III. RESULTS

		Accuracy	Precision	Recall	F1-Score	Hiperparámetros
ViT	**SVM**	**0.9838**	**0.9691**	**1.000**	**0.9843**	**C: 0.1; Kernel: linear; Gamma: scale**
	RF	0.9297	0.9263	0.9362	0.9312	N_Est: 300; Depth: None; MinSamples: 2
	KNN	0.8757	0.8515	0.9149	0.8821	K: 15; Weights: distance; Metric: cosine
PiT	SVM	0.6054	0.6105	0.6170	0.6138	C: 0.1; Kernel: linear; Gamma: scale
	RF	0.6108	0.6196	0.6064	0.6129	N_Est: 100; Depth: 20; MinSamples: 2
	KNN	0.6000	0.5909	0.6915	0.6370	K: 25; Weights: distance; Metric: cosine
CNN	SVM	0.9459	0.9286	0.9681	0.9479	C: 1; Kernel: rbf; Gamma:scale
	RF	0.9405	0.9109	0.9787	0.9436	N_Est: 300; Depth: None; MinSamples: 1
	KNN	0.9459	0.9468	0.9468	0.9468	K: 15; Weights: distance; Metric: cosine
MobileNetV2	SVM	0.6270	0.6404	0.6064	0.6230	C: 1; Kernel: rbf; Gamma: 0.01
	RF	0.5622	0.5684	0.5745	0.5714	N_Est: 300; Depth: None; MinSamples: 1
	KNN	0.5568	0.5625	0.5745	0.5684	K: 15; Weights: distance; Metric: cosine
BotNet	SVM	0.5946	0.6234	0.5106	0.5614	C: 0.1; Kernel: linear; Gamma: scale
	RF	0.6270	0.6344	0.6277	0.6310	N_Est: 200; Depth: 20; MinSamples: 2
	KNN	0.6432	0.6628	0.6064	0.6333	K: 15; Weights: distance; Metric: euclidean
MLPMixer	SVM	0.6486	0.6593	0.6383	0.6486	C: 1; Kernel: rbf; Gamma: scale
	RF	0.6757	0.6771	0.6915	0.6486	N_Est: 300; Depth: None; MinSamples: 2
	KNN	0.6541	0.6630	0.6489	0.6559	K: 15; Weights: distance; Metric: cosine

RESULTS OF THE BEST HYBRID MODEL

Hyperparameters: Linear SVM | C=0.1 | Kernel= Linear | Gamma= Scale

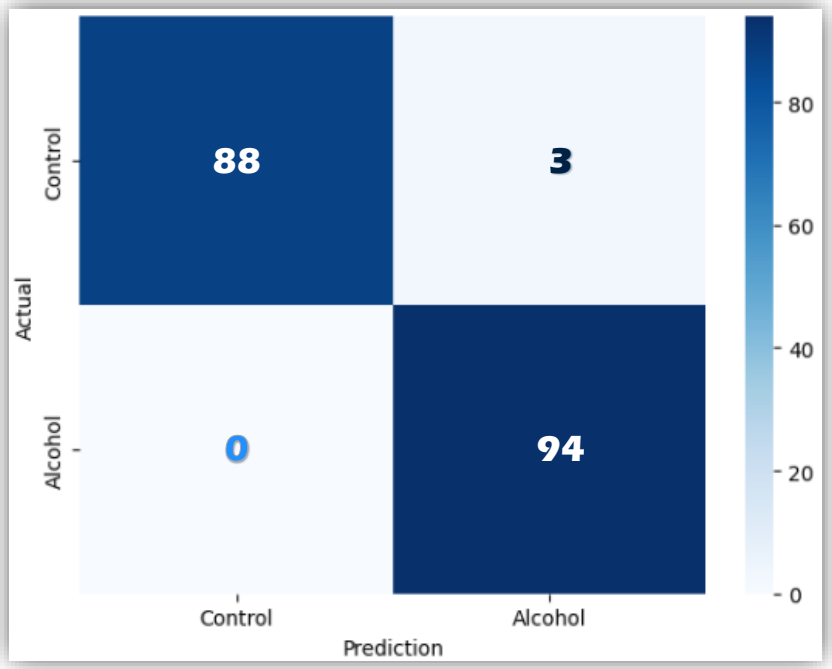
ViT + SVM

98,38% Accuracy	**100%** Recall
96,91% Precision	**98,43%** F1-Score

Key fact: 100% sensitivity means that the model did not miss ANY positive cases of alcoholism.

III. RESULTS

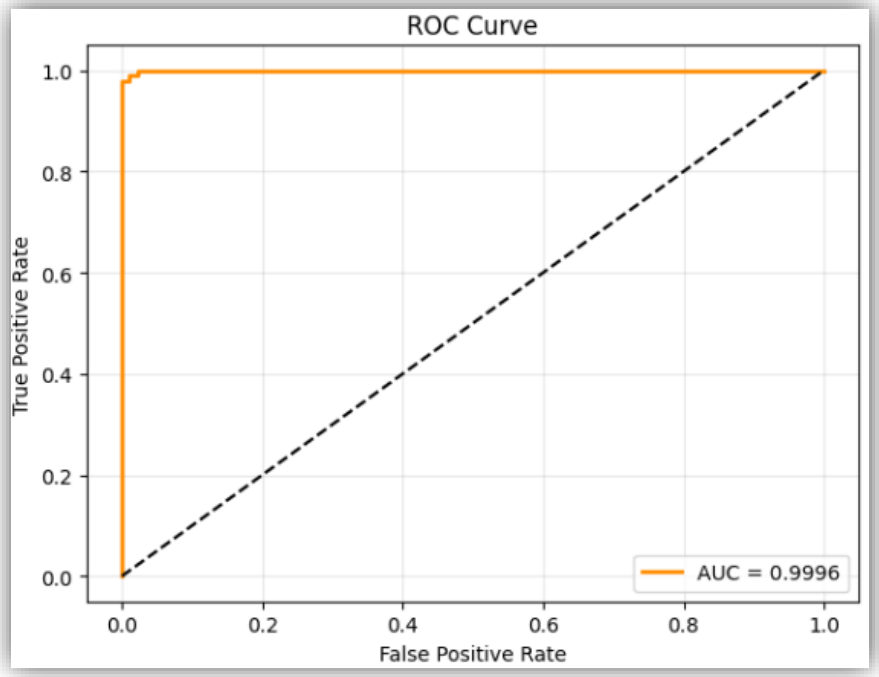
CONFUSION MATRIX



- **88** true negatives (correct controls)
- **3** false positives (minimal alerts)
- **0** false negatives (no alcoholics missed)
- **94** true positives (accurate detection)

ROC CURVE

The ROC curve shows an AUC of 0.9996 with a steep upward slope, accurately identifying actual cases of alcoholism while keeping false positives to a minimum.



IV. CONCLUSIONS

- The study demonstrated the effectiveness of implementing hybrid models for the detection of alcoholism based on EEG signals, combining deep learning and machine learning techniques.

- An EEG dataset on alcoholism in the ICU was compiled, comprising a total of 924 images, including 468 images of individuals with alcoholism and 456 control images.

- Preprocessing metrics such as resizing, normalization, labeling, and the creation of test data were applied to ensure proper preprocessing of the EEG scalogram dataset.

- Six deep learning models and three machine learning models were implemented. They were run with and without hyperparameter tuning.

- It is concluded that the proposed ViT + SVM hybrid model is the most effective, achieving an accuracy of 98.38%, a precision of 96.91%, a recall of 100%, an F1-score of 98.43%, and an AUC of 0.99 on the ROC curve. This demonstrates that it is a highly reliable tool for the early diagnosis of alcoholism.

Thanks for your attention